

Name: _____ Date: _____

Period: _____

Parametrically Defined Circles and Lines Practice — Key

- Fountain: radius 6, center $(-1, 3)$, counterclockwise from $(5, 3)$.
 - $x(t) = -1 + 6 \cos t$, $y(t) = 3 + 6 \sin t$, $0 \leq t \leq 2\pi$
 - At $t = \pi/2$: $x = -1 + 6(0) = -1$, $y = 3 + 6(1) = 9$. Position: $(-1, 9)$.
At $t = \pi$: $x = -1 + 6(-1) = -7$, $y = 3 + 6(0) = 3$. Position: $(-7, 3)$.
- Circle: $x(t) = -2 + 5 \cos t$, $y(t) = 4 + 5 \sin t$.
 - Center $(-2, 4)$, radius 5.**
 - $x(t) = -2$: $-2 + 5 \cos t = -2 \Rightarrow \cos t = 0 \Rightarrow t = \pi/2$ or $t = 3\pi/2$.
At $t = \pi/2$: $y = 4 + 5(1) = 9$. Point: $(-2, 9)$.
At $t = 3\pi/2$: $y = 4 + 5(-1) = -1$. Point: $(-2, -1)$.
- Clockwise, radius 7, center $(2, -3)$ (replace $\sin t$ with $-\sin t$):
 $x(t) = 2 + 7 \cos t$, $y(t) = -3 - 7 \sin t$, $0 \leq t \leq 2\pi$
At $t = \pi/2$: $x = 2 + 7(0) = 2$, $y = -3 - 7(1) = -10$. Position: $(2, -10)$.
- Hiker: $A = (1, 4)$ to $B = (9, 0)$.
 - $x(t) = 1 + 8t$, $y(t) = 4 - 4t$, $0 \leq t \leq 1$
 - At $t = 0.25$: $x = 1 + 2 = 3$, $y = 4 - 1 = 3$. Position: $(3, 3)$.
At $t = 0.75$: $x = 1 + 6 = 7$, $y = 4 - 3 = 1$. Position: $(7, 1)$.
 - $y(t) = 0$: $4 - 4t = 0 \Rightarrow t = 1$. The path reaches $y = 0$ at the endpoint $t = 1$, position $(9, 0)$.

AP-Style Practice — Key

- $x(t) = 3 + 4 \cos t = 3 \Rightarrow 4 \cos t = 0 \Rightarrow \cos t = 0$, so $t = \pi/2$ or $t = 3\pi/2$.
At $t = \pi/2$: $y = -2 + 4(1) = 2$. At $t = 3\pi/2$: $y = -2 + 4(-1) = -6$.
Both values are attained, so the correct answer is **(C)**.
(C) 2 and -6 ← correct
 - (A) 2 — only one of the two values; incomplete.
 - (B) -6 — only one of the two values; incomplete.
 - (D) -2 — that is the y -coordinate of the center, not a point where $x = 3$.

Teacher Note

The AP item tests whether students recognize that $x(t) = 3$ is solved by $\cos t = 0$, which has *two* solutions in $[0, 2\pi)$. Students who report only one value likely stopped after finding $t = \pi/2$. Emphasize: always check for all solutions within the domain.

Extension optional

Extension (optional) — Key:

From $x(t) = h + r \cos t$: $\cos t = (x - h)/r$.

From $y(t) = k + r \sin t$: $\sin t = (y - k)/r$.

Use $\cos^2 t + \sin^2 t = 1$: $\left(\frac{x-h}{r}\right)^2 + \left(\frac{y-k}{r}\right)^2 = 1$, so $(x-h)^2 + (y-k)^2 = r^2$.